

Assessing the suitability and key limitations of Rpath, a complementary R package for Ecopath with Ecosim, for ecosystem modelling.

Ryley Delaney (1103282)

Abstract:

Rpath is an R package complementary to the *Ecopath with Ecosim* application suite used for conducting ecosystem-based management. Here we assess the capabilities of Rpath in reference to Ecopath with Ecosim for potential stakeholders, with a focus on showcasing the potential uses of Rpath, identifying limitations of the package, and identifying potential data sources for interested parties. Rpath is also examined as a potential modelling tool for analysing ecosystem perturbations under different climate scenarios. Results of this investigation found that unstable modelling outcomes breaking fundamental assumptions of the model can be created easily without proper care. Other findings are that while Rpath is highly accessible, allowing for reproducible models to be completed in only a few lines of code, it requires foundational understanding of the modelling framework to defensibly utilise, else unstable models with unrealistic parameters are produced. There was also found to be a notable lack of available data accessibility for modelling, and limited Rpath-specific documentation for teaching the package. Because Rpath documentation is limited and requires bridging from Ecopath with Ecosim, recommendations to interested parties are to implement Ecopath with Ecosim to their modelling pipeline first and then later integrate Rpath if they determine that the additional benefits of integration outweigh the costs.

1. Introduction

Modern fisheries management has been practised since the early 1940's and developed with a strong focus on sustaining fishing activity and resources (Chavance et al., 2004). Despite this, degradation and habitat loss due to poor or nonexistent monitoring, inadequate management-oriented research, and a lack of a coherent aquatic policy are still frequently described (Garcia, 1992; Sutinen & Soboi, 2003). A shift in the paradigm of modern fisheries management has led to steps towards a whole-of-ecosystem approach for fisheries and away from single-species models, which, coupled with increased computational power, has allowed for new tools in ecosystem modelling to be developed and used in fisheries management (Plagányi, 2007). Several ecological network models have since been developed in easy-to-use packages to depict energy budgets within a food web for assessing marine resource management (Link, 2010). However, this same accessibility has also led to the production of models of diminished quality (Link, 2010), potentially indicating that there are barriers or limitations to effective modelling use.

Ecopath with Ecosim (Polovina, 1984; Christensen & Pauly, 1992; Walters et al., 1997; Walters et al., 2000; hereafter EwE) is an ecosystem modelling software suite that acts as a toolbox for exploring and answering ecological questions about energy flows within ecosystems. The suite follows the Ecopath modelling framework, a series of equations that relate energy flows within a system, that was developed based on research conducted in a remote coral reef system of the Hawaiian Islands by Polovina (1984) to quantify the primary production required to support the ecosystem. EwE and its original modelling framework continue to grow in popularity and use (Link, 2010), having been cited in over 2100 publications in the Google Scholar database. This popularity has led to continued development and maintenance, including the framework being applied to both aquatic and terrestrial ecosystems (Fretzer, 2014). Today, it is represented by three primary components: *Ecopath*, *Ecosim*, and *Ecospace*. Simplified, *Ecopath* represents a temporal snapshot of the energy flows within a system, while *Ecosim* and *Ecospace*, represent, respectively, time-dynamic and spatial modalities of analysis. The apparent simplicity of its use through the development of a convenient graphical user interface has greatly expanded EwE's accessibility; however, this is offset by a lack of formulaic reproducibility of any conducted

modelling. Some capabilities and published uses of EwE modelling are described in its best practices (Heymans et al., 2016) including: identification and quantification of major energy flows within an ecosystem, describing ecosystem resources and their interactions among species, evaluating the ecosystem effects of fishing or environmental changes, exploring management policy options through fisheries, evaluating placement and impact of marine protected areas, and testing ecosystem theories on resilience, stability and regime shifts.

The statistical language R, an open-source programming language is familiar to ecologists and of growing popularity, with R or one of its many packages cited in more than one in 100 scholarly articles in agricultural and environmental sciences within Elsevier's Scopus database (Tippmann, 2015). Rpath (Aydin et al., 2025) is introduced as an implementation of EwE that complements the existing software suite (Lucey et al., 2020) and makes use of the extensive graphical capabilities of the R programming language while allowing for data to be imported directly from EwE. Notably, Rpath addresses EwE's problem of reproducibility, with modelling information stored in an easy-to-share script format. Currently, Rpath implements the mass-balance modelling of Polovina (1984), the biomass dynamics methods of Walters et al. (1997), and the life history stanzas methods of Walters et al. (2000). These implementations allow Rpath to analyse changes in energy flow for the purposes of answering ecological questions regarding ecosystem disturbances.

In recent years, conservation has become a key focus within ecology. Anthropogenic climate change has taken centre stage with major consequences expected from the tropicalisation of ecosystems (Vergés et al., 2014). The resulting 'catastrophic shifts' to habitats within marine ecosystems are expected to have strong impacts on biodiversity and abundance through changes in energy flow up the food web (Ling, 2008). This study aimed to investigate Rpath as a potential marine resource management tool for analysing changes in ecosystems from climate change, more explicitly this project set out to: determine the capabilities of Rpath and showcase its potential, particularly in relation to the above scenario; investigate sources for teaching EwE and Rpath; determine what data is necessary for trophic web modelling and potential data sources; and identify key barriers and limitations to the use of EwE and Rpath for potential stakeholders.

2. Methods

2.1 Model Equations

The EwE modelling framework relates energy flows with biomass to estimate values within a trophic web, known as ‘mass-balancing’ (Christensen & Walters, 2004). It operates based on two master equations that define relationships between the production of functional groups and the consumption of the functional groups that predate them within a ‘box’ ecosystem. A box ecosystem is an ecosystem made up of functional groups or species, known as ‘boxes’ where each box must satisfy the condition that production equals predation, mortality, and export (Christensen & Walters, 2024). The mass-balancing of an ecosystem requires three out of the possible four master equation parameters for each functional group to develop and estimate the fourth. These parameters are: biomass, the ratio of production to biomass, the ratio of consumption to biomass, and the ecotrophic efficiency, which represents the proportion of the energy produced within the ecosystem that is also used within the ecosystem. In Rpath, several other ancillary parameters will also be calculated, including the biomass and ecotrophic efficiency of detritus groups, trophic levels, and the mortality terms of functional groups (Lucey et al., 2020). Other required model inputs include: a diet matrix, consisting of the proportion of prey that make up the diet of each functional group; and fleets representing fishing removals and discards. Removals and master equation parameters are represented in weight per unit area per unit time, but most commonly in tonnes per square kilometre per year.

2.2 Models and Ensembles:

For this report, two artifacts were created to investigate and showcase Rpath’s capabilities: a mass-balanced iterative model built from data sourced from expert opinion supplemented by a published EwE model of the Kimberley region in Western Australia (Boschetti et al., 2017); and an ensemble of dynamic simulations under different climate scenarios applied to the parameters of Anchovy Bay, a set of example model parameters included within Rpath. Data inputs for the final iterative model and diet matrix are included in Tables 1 and 2. Sources for main input parameters are included in Appendix 1. All diet information was sourced from expert opinion on marine ecosystems.

Table 1: Input parameters for the Rpath model. Values in italics are calculated by Rpath.

Group	Trophic Level	B	PB	QB	EE	Bioacc	Unassim	Fleet Removals	Fleet Discards
Epibenthos	2.23	106.7044	2.9	10	0.95	0	0.2	0	0
Luderick	2.74	16.0329	0.78	11.63	0.95	0	0.2	0.0125	0
Black Rabbitfish	2.49	19.43	2.15	9.7	0.95	0	0.2	0.125	0
Yellowtail Kingfish	2.63	3.355	0.54	23.5	0.083	0	0.2	0.15	0
Ornate Wobbegong	2.63	0.0728	0.33	3.8	0.52	0	0.2	0.0125	0
Zooplankton	2.11	36.54	55	70	0.91	0	0.2	0	0
Phytoplankton	1.0	41.87	90	NA	0.99	0	0	0	0
Macrophytes	1.0	2.51	24	NA	0.95	0	0	0	0
Detritus	1.0	2.51	24	NA	0.95	0	0	0	0
Fleet	1.0	20	NA	NA	NA	0	0	0	0
	3.58	NA	NA	NA	NA	NA	NA	NA	NA

Table 2: Diet matrix for final iterative model. Primary producers' columns have been excluded.

Group	Epibenthos	Luderick	Black Rabbitfish	Yellowtail Kingfish	Ornate Wobbegong	Zooplankton
Epibenthos	0.1	0.6	0.4	NA	NA	NA
Luderick	NA	NA	NA	0.15	0.15	NA
Black Rabbitfish	NA	NA	NA	0.25	0.25	NA
Yellowtail Kingfish	NA	NA	NA	NA	NA	NA
Ornate Wobbegong	NA	NA	NA	NA	NA	NA
Zooplankton	0.1	NA	NA	NA	NA	0.1
Phytoplankton	0.1	NA	NA	NA	NA	0.8
Macrophytes	0.1	0.4	0.4	NA	NA	NA
Detritus	0.6	NA	0.2	0.6	0.6	0.1
Import	NA	NA	NA	NA	NA	NA

3. Results

3.1 Iterative Model:

The final iterative modelled ecosystem after mass-balancing (Figure 1) represents a trophic web consisting of the functional groups: benthos (*Ecklonia radiata* and *Sargassum sp.*), representing detrital algae; black rabbitfish (*Siganus Fuscescens*), representing the targeted herbivore; luderick (*Girella tricuspidata*), representing the non-targeted herbivore; yellowtail kingfish (*Seriola lalandi*), representing the targeted carnivore; and the ornate wobbegong (*Orectolobus ornatus*), representing the non-targeted carnivore. The ecosystem contains no apex predator, resulting in limited energy production from carnivorous groups being used within the system, represented by the low ecotrophic efficiency and biomass values of carnivorous groups calculated by the model.

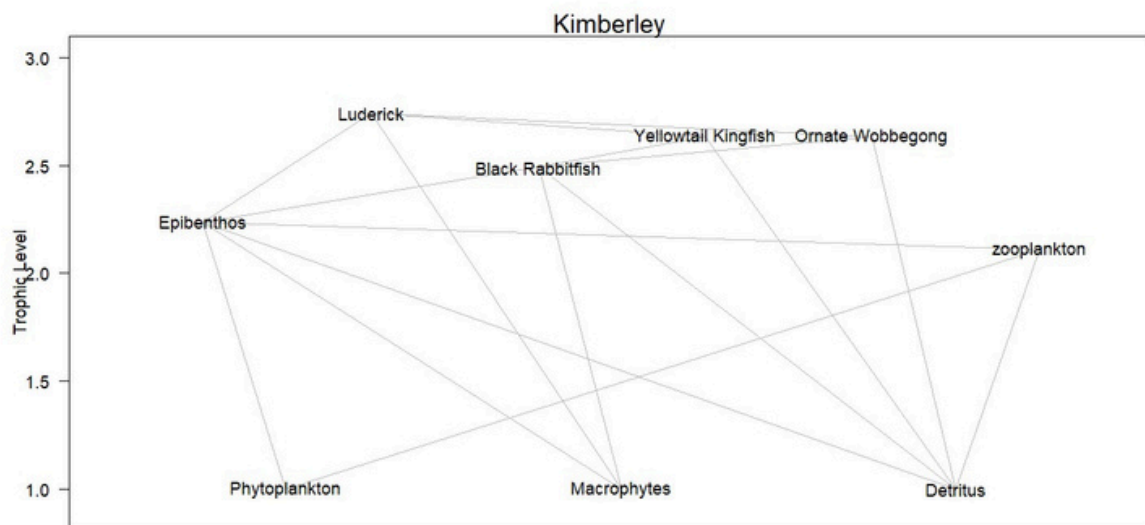


Figure 1: The output trophic food web model calculated by Rpath. The model places luderick at the highest trophic level (2.74) within the web, including above its predators.

A dynamic simulation of the model over 100 years (Figure 2) results in the relative biomass values of several functional groups booming, and one of the two primary producers, macrophytes, crashing.

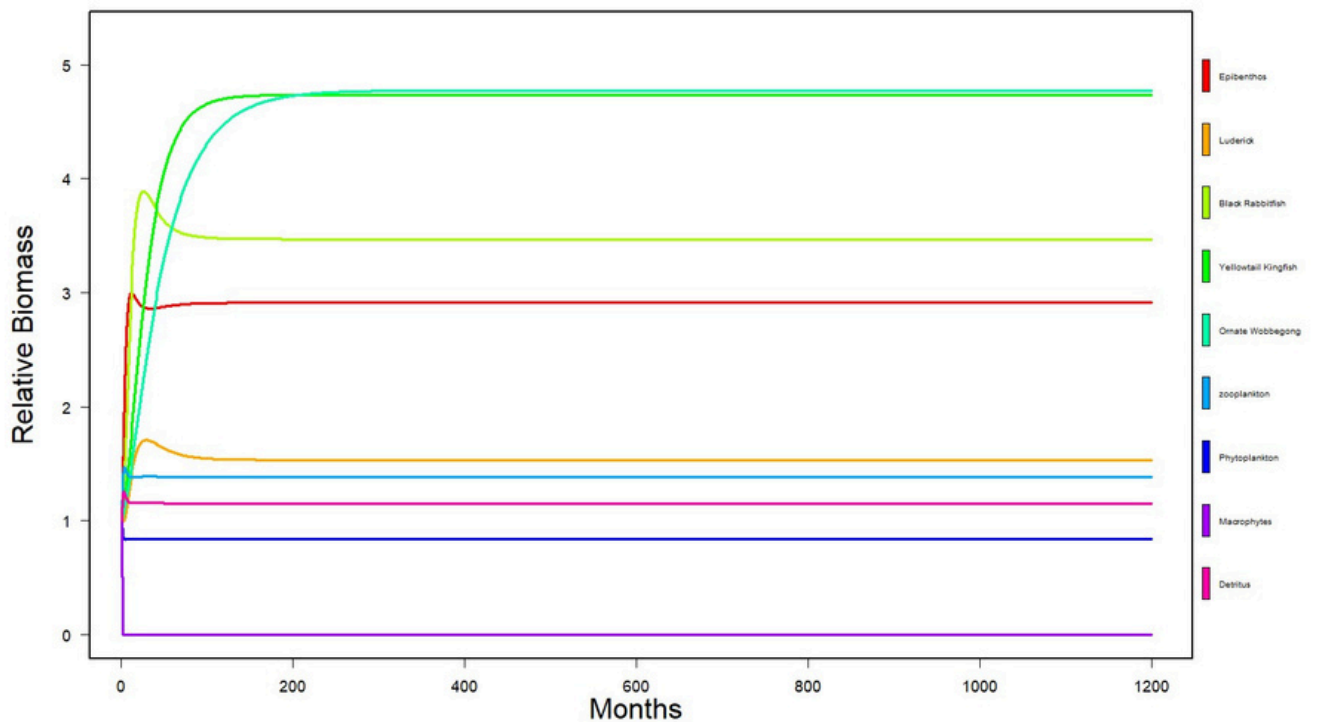


Figure 2: Relative biomass values of functional groups over 100 years. Macrophyte biomass crashes within the first year while epibenthos, black rabbitfish, ornate wobbegong and yellowtail kingfish all boom.

3.2 Dynamic Simulation Ensembles:

With a valid ecosystem model, an ensemble of dynamic simulations can be run and plotted to examine the effects of different climate scenarios on an ecosystem within a single R script. Dynamic simulations on the Anchovy Bay model parameters displayed increasingly severe, non-linear, fluctuations in relative biomass values as environmental forcing was applied. Simulations were conducted on a steady-state model under five scenarios: no perturbations leading to no biomass changes, minor perturbations causing the biomass of benthos to decrease by 3%, moderate perturbations causing benthos to decrease by 5% and cod to increase by 3%, extreme perturbations causing benthos to decrease by 7% and of cod to increase by 5%, and catastrophic perturbations causing the benthos to decrease by 10% and cod to increase by 7%.

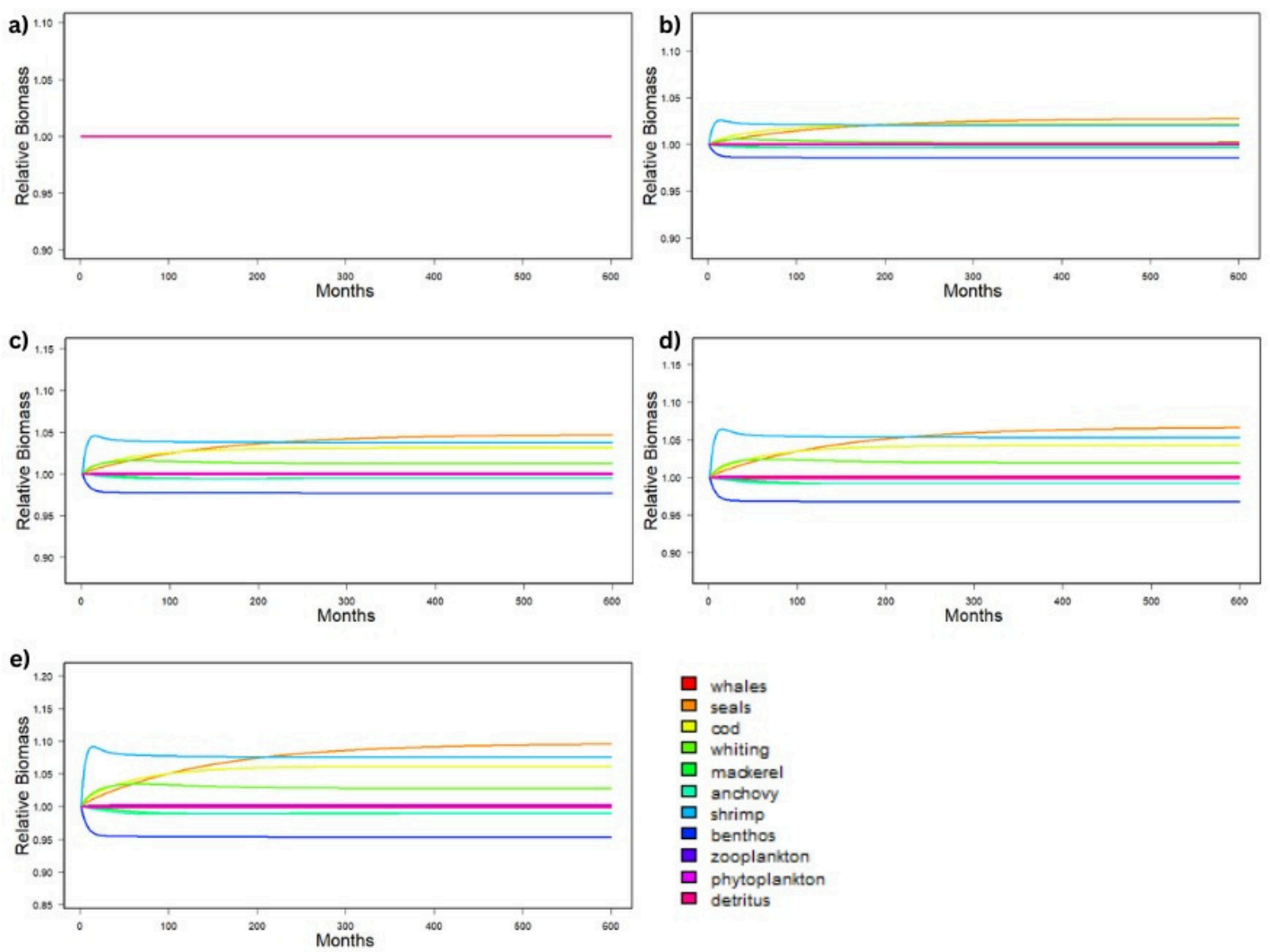


Figure 3: A series of dynamic simulations run over 50 years using the Anchovy Bay model parameters built into Rpath. a) Scenario 1: no perturbations. b) Scenario 2: minor perturbations. c) Scenario 3: moderate perturbations, d) Scenario 4: extreme perturbations. e) Scenario 5: catastrophic perturbations.

4. Discussion

Before the implementation of a modelling tool such as Rpath or the EwE framework, it is necessary to have an understanding of the problem being investigated and ensure alignment with the potential insight that can be gained from a given modelling framework. In this case, Rpath claims an accessible, robust and powerful solution to ecological problems of varying contexts, asserting itself with several publications as evidence of its flexibility. Any modelling solution intended to aid in decision-making requires for the modelling product to be defensible in a decision-making context, the creation of a defensible modelling product requires: an understanding of the necessary inputs or data required for the model, access to appropriate data sources, and an understanding of the capabilities and limitations of the product within the specific context of the problem. Its also worth considering whether the modelling product can still provide insight within its limitations. Here we interpret the products of this modelling framework investigation and outline key insights gained for potential stakeholders interested in Rpath for answering ecological questions.

4.1 Model Interpretation:

When failing to mass-balance models in Rpath, the console will suggest reasons for mass-balancing failure. For example, excess energy within the model must flow to an energy sink, represented by a detrital group. In the case of a mass-balance failure, the algorithm might report, among other diagnostic criteria, that a detrital group is missing. It is worth noting however, that Rpath will violate underlying assumptions and logical constraints in order to complete mass-balancing, therefore it is the responsibility of the user to test underlying model assumptions and constraints. The iterative model, breaks several ecological assumptions and logical constraints, for example, that species predators must be of a higher trophic level than their prey, and have a lower biomass value. In our iterative model, Rpath places Luderick (*Girella tricuspidata*) at the highest trophic level within the ecosystem despite having a predator, indicating, in terms of mass-balancing equations, that it is either overproducing or being under consumed by its predators, which determines the trophic level Rpath denotes to a given functional group (Lucey et al., 2020). During the dynamic simulation, extreme changes in relative biomass values occur after the first year, indicating either a volatile ecosystem or, more probably, unrealistic ecosystem parameters.

While logical constraints and assumption tests are described by Link (2010) and Darwall et al. (2010) to ensure model quality, they must still be completed by the user. Furthermore, it is acknowledged that, because EwE models are more accessible compared to other ecosystem modelling techniques, problems with quality assurance have arisen (Heymans et al., 2016). In a meta-analysis conducted by Heymans et al. (2014), it was reported that of 105 models, very few used the prebalancing checks described in the best practices for EwE (Heymans et al., 2016). These 'completed' models that violate thermodynamic principles and are unstable, showcase that Rpath is capable of producing models with indefensible implications for management when used with incomplete data or without care.

4.2 Considerations for Defensible Modelling:

As shown, once appropriate model parameters have been obtained, tested, and validated, investigating ecological questions is relatively straightforward, however, creating robust, generalisable models that maintain validity requires several key components: reasonably complete data capable of answering the given research question (Plagányi, 2007); a series of diagnostic criteria to validate model parameters, and a strong foundational understanding of Rpath and its underlying modelling framework, Ecopath with Ecosim. While EwE's mass balance algorithms were designed to work with data sources available to fisheries (Lucey et al., 2020), specific means of data sourcing are outlined within EwE's best practices documentation (Heymans et al., 2016). Biomass values are recommended to be calculated through standard surveying or stock assessment methods described by Hilborn & Walters (2015) and Walters (2004). For top predators within the model, it is recommended to use high-quality and up-to-date biomass data, preferably census, as they constrain the parameters of other consumers within the ecosystem (Heymans et al., 2016). The best available data source for diet estimates is obtained from stomach content analyses on predators based on weight (Stergiou & Karpouzi, 2002); however, they can also be obtained from isotopic analyses (Ramsvatn, 2013). Production-to-biomass ratios (P/B) and consumption-to-biomass ratios (Q/B) can be obtained from laboratory experiments but when not readily available, P/B can be calculated empirically based on the turnover rate of a given functional group (Equation 1), where it is equal to the total mortality of the group. Total mortality is equal to the fishing mortality, represented by catch over biomass per year and which can be easily obtained from fishery stock assessments or databases such as Qfish (Department of Agriculture and Fisheries, 2026), plus natural mortality, which can be estimated from Pauly's (1980) empirical equation using the total length (cm) or wet weight (g) at infinity for the population, the von Bertalanffy growth parameter per year, and the mean

annual temperature at which the population is maintained in degrees (°C). Parameters to estimate natural mortality and Q/B estimates can be obtained from Fishbase (Froese & Pauly, 2025), but many species datasets contain small sample sizes or require length values to be converted using length-to-length ratios, decreasing generalisation. Q/B values can also be calculated using the empirical ratio of Palomares & Pauly (1998).

Parameters can also be obtained from previously published analogous ecosystem models. Published literature is the most common source for analogous ecosystem parameters; however, investigations into models for this project found that publications often lack enough information for complete model replication, often with incomplete or missing diet matrices. Ecobase is an online repository featuring over 400 EwE models created for researchers to publish and communicate their work (Colléter et al., 2013), with some models including the original EwE suite parameters for download, however the database contains several broken links, or models with incomplete metadata or parameter information for replication. It's also worth noting that, to be used in Rpath, model information obtained from Ecobase must be downloaded and then converted from the EwE suite to Rpath.

4.3 EwE and Rpath Documentation:

While at the time of writing, Rpath documentation is limited, there is extensive documentation on the EwE suite and its modelling framework. Primary sources include an EwE suite user's guide (Christensen et al., 2023) that includes modelling tutorials and examples of how to obtain data, and a published textbook designed for explaining concepts and teaching a course on Ecopath with Ecosim (Christensen & Walters, 2024), which includes background information and explanations of the underlying framework. Both of these textbooks are freely available online and under a Creative Commons license to incentivise collaboration, including for commercial purposes. The tutorials and information presented in both of these textbooks are not directly transferable to Rpath and require a bridging of information based on the differences in inputs and user interface. To aid in this bridging, Rpath, like all packages in R, contains information on its functions and a series of vignettes that demonstrate the core functionality of the package, however the vignettes are by no means a replacement for dedicated tutorials and documentation that act as a user guide for Rpath or aiding in the transition between Rpath and EwE. It is also worth noting that the Rpath vignettes are not tutorials on the EwE modelling framework within Rpath; they are demonstrations of how to build and run the model, described on the basis that the user has an understanding of what they are doing and why they would be doing it. At the time of

this report, the official Rpath landing page features papers on the use of Rpath for management strategy evaluations (Lucey et al., 2021) and sensitivity analysis (Whitehouse & Aydin, 2020), as well as diagnostics, which have already been discussed, but notably lacks a dedicated Rpath user manual.

4.4 Conclusion:

Rpath is described as a complementary package to the EwE suite, yet it would take much better advantage of the prolific use of R within the ecological sphere as a stand-alone package with a dedicated teaching approach. At present, an ecologist already familiar with R intended to directly apply that knowledge within the EwE framework must first take an indirect path in learning EwE within a separate application and then determine how to bridge that knowledge to what they are familiar with. The development of Rpath was partly incentivised to take advantage of the extensive community and online support available to researchers already using R (Lucey et al., 2020). Therefore, ecologists would be more incentivised to use Rpath if streamlined documentation existed, as they could adapt the skills they already possess, as opposed to being forced to take the intermediary step of learning a new system entirely. This would increase Rpath's usage and lead to extended collaboration and development of the package. At minimum, a dedicated section within the EwE user guide that explains how to use the same functionality of EwE within Rpath would help bridge the gap between the two applications. Furthermore, it would likely affect the quality of future models by creating a stronger foundational understanding of the modelling framework and decreasing the risk of error during data conversion between applications. The lack of a dedicated understanding of any modelling framework leads to misuse during application, a phenomenon seen in multiple ecological contexts. One example is during the use of river classification for land management decisions without strong foundational knowledge of the hydrological principles underlying them, causing poor decision-making outcomes (Doyle et al., 1999). As such, accessibility of the modelling system without data transparency and clearly accessible foundational knowledge lays the groundwork for misuse and management practices that lack defensibility, eventually reducing the credibility of the technique itself.

Rpath is simple to both use and misuse, and while it has strong capabilities for ecological modelling applications, including examining the effects of perturbations during different climate change scenarios, it lacks a dedicated set of documentation or user guides and requires data that, at present, can be difficult to obtain. This report recommends in summary that stakeholders interested in the modelling system learn from the already

established textbooks on EwE and its modelling framework, and then adopt Rpath into their modelling pipeline if they determine that the capabilities provided by R would be worthwhile. Investigation's into Rpath also found that the key barriers and limitations that currently prevent more integrated usage of Rpath and its underlying modelling framework revolve around three core themes: a lack of dedicated teaching and documentation on the framework, open-source models and limited data accessibility, and a lack of integrated diagnostic criteria built into the software, all of which should be carefully considered before modelintegration.

References:

- Aydin, K., Lucey, S., Gaichas, S., Weisberg, S., & Whitehouse, A. (2025). *Rpath: R implementation of Ecopath with Ecosim*. <https://noaa-edab.github.io/Rpath/>
- Boschetti, F., Lozano-Montes, H., Stelfox, B., Bulman, C., Strzelecki, J., & Hughes, M. (2017). *Knowledge Integration and Management Strategy Evaluation (MSE) Modelling* (No. 2.2.8). Western Australia Marine Science Institution (WAMSI).
- Chavance, P., Ba, M., Gascuel, D., Vakily, J. M., & Pauly, D. (2004). *Pêcheries maritimes, écosystèmes et sociétés en Afrique de l'Ouest: Un demi-siècle de changement: Actes du symposium international = Marine fisheries, ecosystems and societies in West Africa: Half a century of change* (P. Chavance, M. Ba, D. Gascuel, J. M. Vakily, & D. Pauly, Eds.). Office des Publications Officielles des Communautés Européennes. <https://www.documentation.ird.fr/hor/fdi:010056013>
- Christensen, V., & Pauly, D. (1992). ECOPATH II – a software for balancing steady-state ecosystem models and calculating network characteristics. *Ecological Modelling*, 61(3–4), 169–185. [https://doi.org/10.1016/0304-3800\(92\)90016-8](https://doi.org/10.1016/0304-3800(92)90016-8)
- Christensen, V., Steenbeek, J., & Walters, Carl J. (2023). *User Guide for Ecopath with Ecosim (EwE): Vol. 1.1*. Vancouver, BC: The University of British Columbia. <https://pressbooks.bccampus.ca/eweguide/>
- Christensen, V., & Walters, C. J. (2004). Ecopath with Ecosim: Methods, capabilities and limitations. *Ecological Modelling*, 172(2–4), 109–139. <https://doi.org/10.1016/j.ecolmodel.2003.09.003>
- Christensen, V., & Walters, C. J. (2024). *Ecosystem Modelling with Ecosim (EwE)* [Application/pdf,application/epub+zip,text/html]. University of British Columbia. <https://doi.org/10.14288/24D7-AB68>
- Colléter, M., Valls, A., Guitton, J., Lyne, M., Arreguín- Sánchez, F., Christensen, V., Gascuel, D. D., & Pauly, D. (2013). *EcoBase: A repository solution to gather and communicate*

information from EwE models (Research Report No. Fisheries Centre Research Report 21(1).; p. 60 pages). University of British Columbia.

<https://institut-agro-rennes-angers.hal.science/hal-01103442>

Darwall, W. R. T., Allison, E. H., Turner, G. F., & Irvine, K. (2010). Lake of flies, or lake of fish? A trophic model of Lake Malawi. *Ecological Modelling*, 221(4), 713–727.

<https://doi.org/10.1016/j.ecolmodel.2009.11.001>

Doyle, M. W., Miller, D. E., & Harbor, J. M. (1999). Should river restoration be based on classification schemes or process models? Insights from the history of geomorphology. *ASCE International Conference on Water Resources Engineering, Seattle, Washington*, 1–9.

Fretzer, S. (2014). IMPLEMENTING THE HABITATS DIRECTIVE IN GERMANY: CASE CONVENTIONS VERSUS ECOPATH, ECOSIM & ECOSPACE 27. *Fisheries Centre Research Reports 2014 Volume 22 Number 3*, 22(3), 64.

Froese, R., & Pauly, D. (2025). *Fishbase* (Version 11/2025) [Dataset].

Garcia, S. M. (1992). Ocean fisheries management. The FAO programme. *Ocean Management in Global Change*.

Heymans, J. J., Coll, M., Libralato, S., Morissette, L., & Christensen, V. (2014). Global Patterns in Ecological Indicators of Marine Food Webs: A Modelling Approach. *PLoS ONE*, 9(4), e95845. <https://doi.org/10.1371/journal.pone.0095845>

Heymans, J. J., Coll, M., Link, J. S., Mackinson, S., Steenbeek, J., Walters, C., & Christensen, V. (2016). Best practice in Ecopath with Ecosim food-web models for ecosystem-based management. *Ecological Modelling*, 331, 173–184.

<https://doi.org/10.1016/j.ecolmodel.2015.12.007>

Hilborn, R., & Walters, C. J. (2015). *Quantitative fisheries stock assessment: Choice, dynamics and uncertainty* (Digital duplication of the orig. ed. 1992). Springer.

Larkum, A. W. D. (1986). A study of growth and primary production in Eckloniaradiat (C.Ag.) J. Agardh (Laminariales) at a sheltered site in Port Jackson, New South Wales. *Journal of Experimental Marine Biology and Ecology*, 96(2), 177–190.

[https://doi.org/10.1016/0022-0981\(86\)90241-8](https://doi.org/10.1016/0022-0981(86)90241-8)

Ling, S. D. (2008). Range expansion of a habitat-modifying species leads to loss of taxonomic diversity: A new and impoverished reef state. *Oecologia*, 156(4),

883–894. <https://doi.org/10.1007/s00442-008-1043-9>

Link, J. S. (2010). Adding rigor to ecological network models by evaluating a set of pre-balance diagnostics: A plea for PREBAL. *Ecological Modelling*, 221(12),

1580–1591. <https://doi.org/10.1016/j.ecolmodel.2010.03.012>

Lucey, S. M., Aydin, K. Y., Gaichas, S. K., Cadrin, S. X., Fay, G., Fogarty, M. J., & Punt, A.

(2021). Evaluating fishery management strategies using an ecosystem model as an operating model. *Fisheries Research*, 234, 105780.

<https://doi.org/10.1016/j.fishres.2020.105780>

Lucey, S. M., Gaichas, S. K., & Aydin, K. Y. (2020). Conducting reproducible ecosystem modeling using the open source mass balance model Rpath. *Ecological Modelling*,

427, 109057. <https://doi.org/10.1016/j.ecolmodel.2020.109057>

Palomares, M. L. D., & Pauly, D. (1998). Predicting food consumption of fish populations as functions of mortality, food type, morphometrics, temperature and salinity. *Marine and*

Freshwater Research, 49(5), 447–453. <https://doi.org/10.1071/MF98015> Pauly, D. (1980).

On the interrelationships between natural mortality, growth parameters,

and mean environmental temperature in 175 fish stocks. *ICES Journal of Marine Science*, 39(2), 175–192.

Plagányi, É. E. (2007). *Models for an ecosystem approach to fisheries*. Food and Agriculture Organization of the United Nations.

Polovina, J. J. (1984). Model of a coral reef ecosystem: I. The ECOPATH model and its application to French Frigate Shoals. *Coral Reefs*, 3(1), 1–11.

<https://doi.org/10.1007/BF00306135>

Queensland Government, Department of Agriculture and Fisheries. (2026). *Qfish* [Dataset]. <http://qfish.fisheries.qld.gov.au>

Ramsvatn, S. (2013). *Investigating coastal ecosystem structure and dynamics using Ecopath mass-balance modelling and stable isotope data- A dissertation for the degree of Philosophiae Doctor*.

Stergiou, K. I., & Karpouzi, V. S. (2002). Feeding habits and trophic levels of Mediterranean fish. *Reviews in Fish Biology and Fisheries*, 11(3), 217–254.

<https://doi.org/10.1023/A:1020556722822>

Sutinen, J. G., & Soboil, M. (2003). The performance of fisheries management systems and the ecosystem challeng. *Responsible Fisheries in the Marine Ecosystem*.

Tippmann, S. (2015). Programming tools: Adventures with R. *Nature*, 517(7532), 109–110. <https://doi.org/10.1038/517109a>

Vergés, A., Steinberg, P. D., Hay, M. E., Poore, A. G. B., Campbell, A. H., Ballesteros, E., Heck, K. L., Booth, D. J., Coleman, M. A., Feary, D. A., Figueira, W., Langlois, T., Marzinelli, E. M., Mizerek, T., Mumby, P. J., Nakamura, Y., Roughan, M., Van Sebille, E., Gupta, A. S., ... Wilson, S. K. (2014). The tropicalization of temperate marine ecosystems: Climate-mediated changes in herbivory and community phase shifts. *Proceedings of the Royal Society B: Biological Sciences*, 281(1789), 20140846.

<https://doi.org/10.1098/rspb.2014.0846>

Walters, C., Christensen, V., & Pauly, D. (1997). Structuring dynamic models of exploited ecosystems from trophic mass-balance assessments. *Reviews in Fish Biology and Fisheries*, 7(2), 139–172. <https://doi.org/10.1023/A:1018479526149>

Walters, C. J. (with Martell, S. J. D.). (2004). *Fisheries Ecology and Management*.
Princeton University Press.

Walters, C., Pauly, D., Christensen, V., & Kitchell, J. F. (2000). Representing Density
Dependent Consequences of Life History Strategies in Aquatic Ecosystems:
EcoSim II. *Ecosystems*, 3(1), 70–83. <https://doi.org/10.1007/s100210000011>

Whitehouse, G. A., & Aydin, K. Y. (2020). Assessing the sensitivity of three Alaska marine
food webs to perturbations: An example of Ecosim simulations using Rpath.
Ecological Modelling, 429, 109074.
<https://doi.org/10.1016/j.ecolmodel.2020.109074>

Appendices:

Appendix 1: Final Iterative Model Input Values Sources:

Group	Biomass	PB	QB	EE	Bioacc	Unassim	Fleet	Fleet Discards
Epibenthos	<i>Rpath</i>	(Larkum, 1986)	Boschetti	Boschetti	0	0.2	0	0
Luderick	<i>Rpath</i>	Boschetti	Fishbase	Boschetti	0	0.2	EO	0
Black Rabbitfish	Boschetti	Boschetti	Fishbase	Boschetti	0	0.2	EO	0
Yellowtail	Boschetti	Boschetti	Fishbase	<i>Rpath</i>	0	0.2	EO	0
Kingfish Ornate	Boschetti	Boschetti	Fishbase	Boschetti	0	0.2	EO	0
Wobbegong	<i>Rpath</i>	Boschetti	Fishbase	Boschetti	0	0.2	EO	0
Zooplankton	Boschetti	Boschetti	Boschetti	Boschetti	0	0.2	0	0
Phytoplankton	Boschetti	Boschetti	NA	EO	0	0	0	0
Macrophytes	Boschetti	Boschetti	NA	Boschetti	0	0	0	0
Detritus	Boschetti	Boschetti	NA	Boschetti	0	0	0	0
Fleet	20	NA	NA	NA	0	0	0	0
	NA	NA	NA	NA	NA	NA	NA	NA

Note: EO = expert opinion, Boschetti = Boschetti et al., 2017.